

Surname MEMO Van Stud. #. _____

Initials _____ Voorletters Total/Totaal: _____/27

The undersigned hereby confirms that she/he is familiar with and accepts the examination regulations.

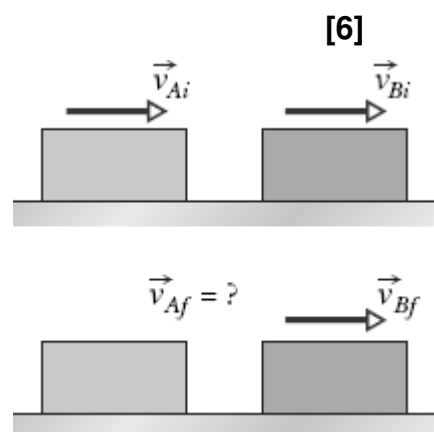
Ondergetekende bevestig hiermee dat hy/sy op hoogte is met die eksamenregulasies en dat hy/sy hom/haar daaraan onderwerp.

Signature _____ Handtekening Date _____ Datum

Question 1/Vraag 1

In the Figure, block A (mass 1.6 kg) slides into block B (mass 2.4 kg), along a frictionless surface. The directions of three velocities before (i) and after (f) the collision are indicated; the corresponding speeds are , $v_{Ai} = 5.5 \text{ m/s}$, $v_{Bi} = 2.5 \text{ m/s}$ and $v_{Bf} = 4.9 \text{ m/s}$.

Die figuur dui blok A (massa 1.6 kg) wat oor 'n wrywinglose oppervlak gly in blok B (massa 2.4 kg) vas. Die rigtings van drie snelhede voor (i) en na (f) die botsing word aangedui; die ooreenstemmende waardes van die drie is $v_{Ai} = 5.5 \text{ m/s}$, $v_{Bi} = 2.5 \text{ m/s}$ en $v_{Bf} = 4.9 \text{ m/s}$.



- a) What are the velocity v_{Af} ?
Wat is die snelheid v_{Af} ?

[3]

$$m_A v_{Ai} + m_B v_{Bi} = m_A v_{Af} + m_B v_{Bf}$$

and

$$\begin{aligned} v_{Af} &= \frac{m_A v_{Ai} + m_B v_{Bi} - m_B v_{Bf}}{m_A} \\ &= \frac{(1.6 \text{ kg})(5.5 \text{ m/s}) + (2.4 \text{ kg})(2.5 \text{ m/s}) - (2.4 \text{ kg})(4.9 \text{ m/s})}{1.6 \text{ kg}} \\ &= 1.9 \text{ m/s.} \end{aligned}$$

The block continues going to the right after the collision.

- b) Is the collision elastic? Give reasons for your answer.
Is hierdie 'n elastiese botsing? Gee redes vir jou antwoord

[3]

To see if the collision is elastic, we compare the total kinetic energy before the collision with the total kinetic energy after the collision. The total kinetic energy before is

$$\begin{aligned} K_i &= \frac{1}{2} m_A v_{Ai}^2 + \frac{1}{2} m_B v_{Bi}^2 \\ &= \frac{1}{2} (1.6 \text{ kg}) (5.5 \text{ m/s})^2 + \frac{1}{2} (2.4 \text{ kg}) (2.5 \text{ m/s})^2 \\ &= 31.7 \text{ J.} \end{aligned}$$

The total kinetic energy after is

$$\begin{aligned}K_f &= \frac{1}{2}m_A v_{Af}^2 + \frac{1}{2}m_B v_{Bf}^2 \\&= \frac{1}{2}(1.6 \text{ kg})(1.9 \text{ m/s})^2 + \frac{1}{2}(2.4 \text{ kg})(4.9 \text{ m/s})^2 \\&= 31.7 \text{ J}.\end{aligned}$$

Since $K_i = K_f$ the collision is found to be elastic.

Question 2/Vraag 2

[5]

A wheel has a constant angular acceleration of 3.0 rad/s^2 . During a certain 4.0 s interval, it turns through an angle of 120 rad .

'n Wiel het 'n konstante hoekversnelling van 3.0 rad/s^2 . Gedurende 'n sekere 4.0 s interval, draai dit deur 'n hoek van 120 rad .

- a) Show clearly that the angular speed of the wheel at the beginning of the 4.0 s interval is 24 rad/s ?

Dui duidelik aan dat die hoeksnelheid van die wiel aan die begin van die 4.0 s interval 24 rad/s was?

[3]

We take $t = 0$ at the start of the interval and take the sense of rotation as positive. Then at the end of the interval with $\alpha = 3.0 \text{ rad/s}^2$

$t = 4.0 \text{ s}$ then $\theta = 120 \text{ rad}$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

Then

$$\omega_0 = \frac{\theta - \frac{1}{2}\alpha t^2}{t} = \frac{120 \text{ rad} - \frac{1}{2}(3.0 \text{ rad/s}^2)(4.0 \text{ s})^2}{4.0 \text{ s}} = 24 \text{ rad/s}.$$

- b) What is the linear speed of a point at the rim of the wheel at the beginning of the 4.0 s interval if the radius is 5 cm ?

Wat is die liniêre spoed van 'n punt op die rant van die wiel aan die begin van die 4.0 s interval indien die radius 5 cm is?

[2]

$$v = \omega r$$

$$= (24)(5 \times 10^{-2}) = 1.2 \text{ m/s}$$

Question 3/Vraag 3

[8]

The figure shows a uniform disk that can rotate around its center like a merry-go-round. The disk has a radius of 2.00 cm and a mass of 40.0 grams and is initially at rest.

Starting at time $t = 0$, two forces are to be applied tangentially to the rim as indicated, so that at time $t = 1.25 \text{ s}$ the disk has an angular velocity of 250 rad/s

counterclockwise. Force $\vec{F}_1$ has a magnitude of 0.200 N .

Die figuur dui 'n univorme skuiw aan wat om die middelpunt kan roteer (soos 'n mallemeule). Die skuiw het 'n radius van 2.00 cm en 'n massa van 40.0 g en is aanvanklik



met rus. Beginende by $t = 0$, werk twee kragte tangensiaal in op die rant van die skuif soos aangedui, sodat die skuif 'n hoeksnelheid van 250 rad/s antikloksgewys het teen tyd $t = 1.25$ s. Krag $\vec{F}_1$ het 'n grootte van 0.200 N.

- a) What is magnitude of F_2 ?
Wat is die grootte van F_2 ?

[5]

$$\tau_{\text{net}} = RF_2 - RF_1 = I\alpha,$$

$$\omega = \omega_0 + \alpha t$$

$$\alpha = \frac{\omega - \omega_0}{t} = \frac{\omega}{t}$$

For a disk: $I = \frac{1}{2}MR^2$

Solve for F_2 from first equn:

F_2

$$= F_2 = \frac{1}{2}MR \left(\frac{\omega}{t} \right) + F_1 = \frac{1}{2}(0.04)(0.02) \left(\frac{250}{1.25} \right) + 0.200 \text{ N}$$

$$= 0.28 \text{ N}$$

- b) Determine the angular momentum of the disk at time 1.25 s.
Bepaal die hoekmomentum van die disk teen tyd 1.25 s

[3]

$$L = I\omega = \left(\frac{1}{2}MR^2 \right) 250 \text{ rad/s}$$

$$= \left(\frac{1}{2}(40 \times 10^{-3})(2 \times 10^{-2})^2 \right) 250 \text{ rad/s}$$

$$= 0.002 \text{ kg} \cdot \text{m}^2 \text{s}^{-1}$$

Out of the page

Question 4/Vraag 4

[8]

A 0.10 kg block oscillates back and forth along a straight line on a frictionless horizontal surface. Its displacement from the origin is given by $x = (10 \text{ cm}) \cos \left[(10 \text{ rad/s})t + \frac{\pi}{2} \text{ rad} \right]$

'n 0.10 kg blok ossilleer heen en weer in 'n reguitlyn op 'n wrywinglose horisontale oppervlak. Sy verplasing van die oorsprong word gegee deur

$$x = (10 \text{ cm}) \cos \left[(10 \text{ rad/s})t + \frac{\pi}{2} \text{ rad} \right]$$

- a) Determine the frequency of the oscillations.
Bepaal die frekwensie van die ossillasies.

[2]

$$\omega = 10 \text{ rad/s}$$

$$f = \omega / 2\pi = 1.6 \text{ Hz}$$

- b) Determine the maximum speed acquired by the block.
Bepaal die maksimum spoed wat die blok bereik.

[3]

$$v_m = \omega x_m \text{ and } x_m = 10 \text{ cm}$$

then $v_m = (10 \text{ rad/s})(10 \text{ cm}) = 100 \text{ cm/s}$ or 1.0 m/s

- c) What force, applied to the block by the spring, results in the given oscillation?
Watter krag word deur die veer op die blok uitgeoefen om hierdie gegewe ossillasie te veroorsaak?

[3]

$$\omega = \sqrt{\frac{k}{m}} \Rightarrow k = (0.10 \text{ kg})(10 \text{ rad / s})^2 = 10 \text{ N / m}.$$

Thus, Hooke's law gives $F = -kx = -10x$

Enjoy the rest of the day / Geniet die res van die dag